

Software Requirements Specification

for



TORBIDD: Software Procurement Intelligence Platform

Version 1.0

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Revision History

Name	Date	Reason for changes	Version
Peraya Pasin Chayakarn	23/08/2026	Initial draft	0.1
Peraya Pasin Chayakarn	25/08/2026	Revised based on comments from the Professor (Project Owner/Sponsor)	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and nonfunctional requirements for TORBIDD, a web-based platform that discovers, extracts, and structures software-related procurement opportunities published by Thai government agencies. This document translates the project proposal into a formal requirements baseline to guide design, development, testing, and client sign-off. It is intended for the development team, the project client/stakeholders, and QA staff involved in building and validating the system.

1.2 Product Scope

TORBIDD is a centralized web platform that continuously monitors public procurement announcements from the Bangkok Metropolitan Administration (BMA) and other Thai government agencies, identifies which listings are software-related (as opposed to construction, vehicles, or general goods), and converts unstructured Terms of Reference (TOR) PDF documents into structured, searchable opportunity records. By surfacing budget, deadline, and bidder-qualification details that are otherwise buried inside lengthy PDFs, and by providing search, filtering, bookmarking, and alerting tools, TORBIDD lets software companies, freelance developers, and IT consultants track and evaluate procurement opportunities without manually reading every announcement.

The initial release covers the full pipeline from data collection to user-facing analysis, including automated TOR processing, opportunity discovery and comparison tools, and user/admin account management. Excluded features are listed in Section 7.3 (Out-of-Scope Features).

1.3 Document Conventions

This document follows standard SRS formatting conventions. Requirements are identified with unique identifiers in the format:

[Feature]-[Number] (e.g. AUTH-1).

Priority levels follow MoSCoW-style labeling: High (must-have for launch), Medium (should-have), Low (nice-to-have, may slip to a later phase).

- “TOR” refers to Terms of Reference, the official document an agency publishes describing a procurement's scope, budget, and requirements.
- “BMA” refers to the Bangkok Metropolitan Administration.
- Monetary values are expressed in Thai Baht (฿) unless noted otherwise.

2. Overall Description

2.1 Product Perspective

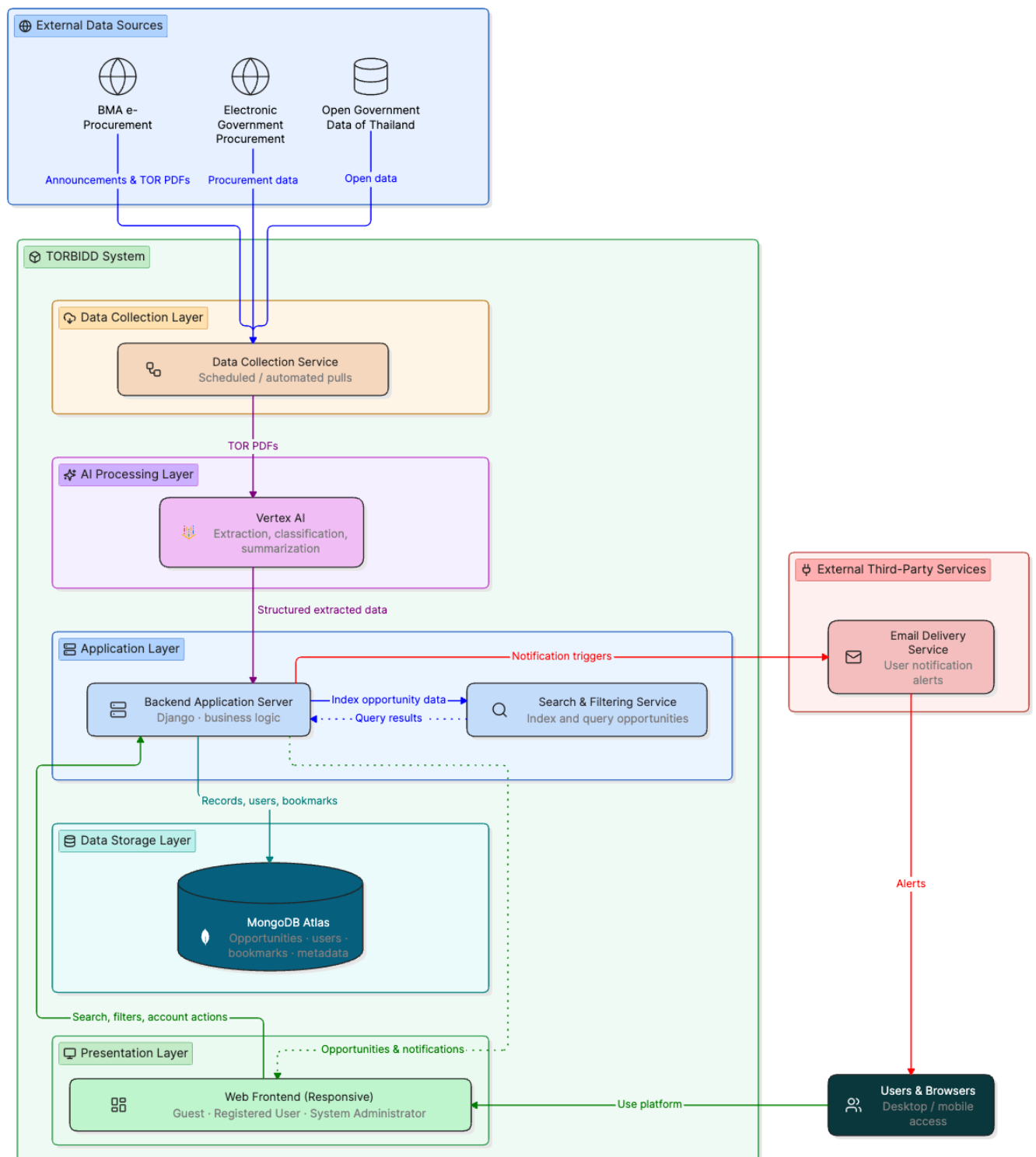
TORBIDD is a new, independent, self-contained web-based product designed to improve access to and interaction with public government procurement information. It is not a component of, or replacement for, any existing government e-procurement system.

TORBIDD acts as an aggregation and intelligence layer that collects and processes publicly available procurement announcements and documents published by government sources (Section 2.6). The system then organizes and re-presents this information in a structured, searchable, and user-friendly format to help users discover and understand relevant procurement opportunities.

The system does not modify, submit information to, or authenticate against government agencies' existing procurement systems. It only consumes publicly available data and documents provided by official government sources.

The system architecture covering data collection, AI-assisted processing, storage, search, and the web frontend is shown in Figure 2.1.

Figure 2.1: TORBIDD System Architecture and Data Flow



2.2 Stakeholders

Stakeholder	Role	Interest	Influence	Needs / Expectations
Project Sponsor / Customer	Initiates and funds the project	High	High	Centralized platform for efficiently discovering and evaluating Thai government software procurement opportunities.
Software Companies	Primary users	High	Medium	Search relevant projects, review qualifications and budgets, compare historical prices, and receive notifications.
Freelance Developers		High	Low	Easily identify relevant government software opportunities without reviewing every TOR.
Business/ Sales Teams		High	Medium	Monitor opportunities, estimate project value, compare historical projects, and prioritize bids.
Government Procurement Departments	Data providers	Low - Medium	High	Provide procurement announcements and TOR documents through government data sources.
Government Project Owners	Procurement owners	Low - Medium	Medium	Provide accurate and complete procurement requirements and TOR documents.
System Administrator	Operates the platform	High	Medium	Manage data synchronization, users, document processing, classification, and system maintenance.
Development Team	Develops the system	High	High	Implement API integration, document extraction, classification, search, normalization, and notification functions.
Government Central API / Data Platform	External data source / integration	Medium	High	Provide standardized procurement data for automated collection and synchronization.

2.3 User Classes and Characteristics

User Class	Frequency	Expertise	Priority	Key Functions
Primary User: Software Studios & Companies	Regular - Frequent	Moderate - High	High	Search, monitor, compare, review pricing
Freelance Developers	Occasional - Regular	High	High	Search, filter by skill/budget, compare
System Integrators / IT Consultants	Regular	Moderate - High	Medium	Compare opportunities, analyze historical pricing
BD / Sales Teams	Regular	Low - Moderate	Medium	Review summaries, budgets, timelines
System Administrators	Frequent	High	High	Monitor ingestion, data, and manage users

2.4 Operating Environment

The TORBIDD system will operate in the following environment

- **Client-side:** Modern desktop and mobile web browsers, including Chrome, Firefox, and other commonly supported browsers. The application will use a responsive layout to support different screen sizes.
- **Server-side:** Cloud-based production environment with the backend deployed using Docker containers.
- **Database:** MongoDB Atlas will be used to store structured opportunity data, including information extracted from tender and opportunity documents.
- **AI and document processing:** Vertex AI will be used for TOR PDF ingestion, document processing, and information extraction.
- **External data sources:** The system will continuously poll or receive data from public government sources, including:
 - Bangkok Metropolitan Administration e-Procurement: egp2.bangkok.go.th
 - Electronic Government Procurement: www.gprocurement.go.th
 - Open Government Data of Thailand: data.go.th
- **Network:** Internet connectivity is required for accessing the web application, government data sources, MongoDB Atlas, Vertex AI, and other required services.
- **Deployment:** The production system will be containerized using Docker to provide a consistent and maintainable deployment environment.

2.5 Design and Implementation Constraints

The design and implementation of TORBIDD are constrained by the following project requirements and limitations

- **CON-1:** The total project budget is limited to ฿1,000,000, as specified in the project proposal.
- **CON-2:** The system must be delivered within a 12-week development period following requirements sign-off.
- **CON-3:** The technology stack (Django backend, MongoDB Atlas, Vertex AI, Docker-based deployment, responsive web interface) is fixed as described in Section 2.5 and constrains implementation choices accordingly.
- **CON-4:** The system shall consume publicly available procurement information and shall not modify, submit data to, or directly manage the source government procurement systems.
- **CON-5:** The system shall support Thai TOR content for document processing and information extraction.
- **CON-6:** The system shall include administrative review and correction capabilities, since AI-based document extraction and classification cannot be assumed to be completely accurate.
- **CON-7:** The system shall be developed within the resources and staffing limitations defined by the project team and schedule.

2.6 Assumptions and Dependencies

2.6.1 Assumptions

The following assumptions are made during the specification and development of TORBIDD. Changes to these assumptions may require corresponding changes to the system or requirements.

- The monitored government sources will continue to publish procurement announcements and TOR documents publicly through web pages, downloadable documents, or public APIs.
- TOR documents are expected to contain Thai text; the document-processing pipeline is assumed to support Thai content.
- Procurement information contained in government announcements and TOR documents is assumed to be sufficiently complete and accurate for extraction by the system.
- Software-related procurement opportunities can be identified and categorized with reasonable accuracy using automated classification, supported by administrator review and correction when necessary.
- Users have a valid email address for receiving system notifications.
- Users have reliable Internet access to use the web application and receive updated procurement information.

2.6.2 Dependencies and Impact of Unavailability

TORBIDD depends on the following external services and infrastructure. The table below summarizes each dependency and the impact if it becomes unavailable.

Dependency	Impact if Unavailable
Government data sources	The affected source's data becomes temporarily unavailable; the corresponding data-collection component may require modification if the source changes structure, format, or access policy.
Vertex AI	AI-based TOR extraction, summarization, and classification may be partially or fully unavailable until service is restored or an alternative solution is implemented.
MongoDB Atlas	Users may be unable to access or update procurement information until the service is restored.
Production hosting infrastructure	The application becomes inaccessible until the hosting environment is restored.
Email delivery service	Procurement alerts and reminders may not be delivered; other core system functions continue to operate normally.

If TOR documents cannot be processed accurately due to changes in document structure, language, or extraction quality, administrator review and manual correction will be required to maintain data quality.

2.7 Requirements Analysis Notes

Open issues, ambiguous or unclear requirements, and justified assumptions identified during requirements analysis are documented in Appendix B.

3. System Features and Use Cases

3.1 Use Case Model

3.1.1 Actors

1. Registered User

- The primary user of TORBIDD.
- Represents software companies, software studios, freelance developers, IT consultants, system integrators, and business development or sales personnel.
- Can search and filter procurement opportunities, view TOR details, bookmark and compare opportunities, configure notifications, and access historical budget information.

2. Guest / Unauthenticated Visitor

- An individual who accesses TORBIDD without signing in.
- Can browse, search, and view publicly available procurement opportunities.
- Must register or sign in before using personalized functions such as bookmarking, comparison, and notifications.

3. System Administrator

- An internal staff member responsible for maintaining the quality and operation of TORBIDD.
- Reviews low-confidence extraction results, corrects extracted fields, changes project classifications, rejects duplicate or irrelevant records, manages users, and monitors data collection.

4. Data Collection Service

- An automated background service that periodically checks monitored government procurement sources.
- Detects new announcements, downloads TOR documents, invokes the extraction and classification process, and stores structured opportunity records.

5. Government Procurement Data Source

- External government websites that publish procurement announcements and TOR documents.
- Initial sources include BMA e-Procurement, gprocurement.go.th, data.go.th, OCSC, and the Ministry of Labour.

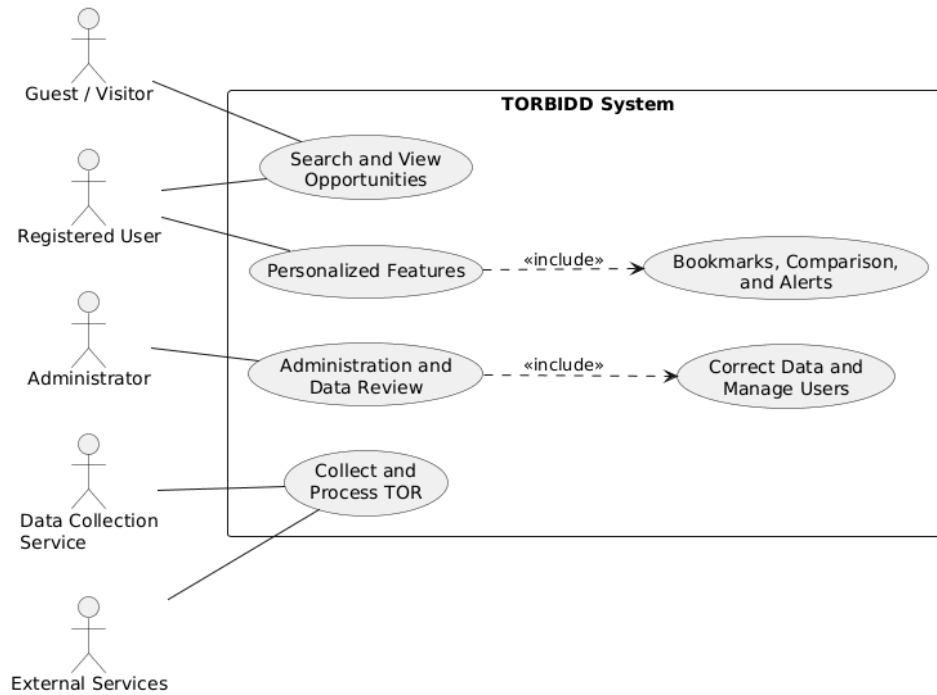
6. Third-Party AI Service

- An external AI service, such as Vertex AI, used to process Thai TOR documents.
- Extracts structured fields and assists in classifying each procurement as a software or non-software project.

7. Email Delivery Service

- An external service responsible for sending account-related emails and procurement opportunity notifications.

3.1.2 System Use Case Diagram



3.1.3 Use Case List

ID	Use Case Name
UC-1	Sign In with Google OAuth
UC-2	View Procurement Opportunity Dashboard
UC-3	Search and Filter Opportunities
UC-4	Extract and Classify TOR Information
UC-5	View Opportunity Details
UC-6	Configure and Receive Opportunity Alerts
UC-7	View Historical Budget and Outlier Analysis
UC-8	Manage User Profile and Preferences
UC-9	Manage Users

Note: The level of detail in user flow diagrams does not necessarily correspond to the textual use case type and is intended to improve understanding of system interactions.

3.2 User Authentication and Access Control

3.2.1 Description and Priority

This feature allows users to securely authenticate using Google Sign-In before accessing any system functionality. All features of the system are restricted to authenticated users.

Priority: Medium

3.2.2 Functional Requirements

- AUTH-1: The system shall allow users to authenticate using Google Sign-In.
- AUTH-2: The system shall restrict access to all personalized system features to authenticated users only such as bookmark, comparison, and notification.
- AUTH-3: The system shall prompt unauthenticated visitors to sign in before accessing personalized content.

3.2.3 Related Use Cases

UC-1: Sign In with Google OAuth

Item	Details
Use Case ID	UC-1
Use Case Name	Sign In with Google OAuth
Primary Actor	Guest / Unauthenticated Visitor
Supporting Actor	Google OAuth Authentication Service
Description	A visitor signs in to TORBIDD using their Google account to access personalized features such as bookmarks, comparisons, saved opportunities, and notifications. If the Google account is not yet associated with a TORBIDD account, the system creates a new TORBIDD account using the information provided by Google.
Preconditions	1. The visitor can access the TORBIDD website. 2. The visitor is not currently authenticated. 3. The visitor has a valid Google account.

Main Flow	1. The visitor opens the TORBIDD sign-in page. 2. The system displays the Sign in with Google button. 3. The visitor selects Sign in with Google. 4. The system redirects the visitor to Google's OAuth authentication page. 5. The visitor authenticates with their Google account and grants the required permissions. 6. Google returns an OAuth authorization result to TORBIDD. 7. The system validates the OAuth result and retrieves the required Google account information. 8. The system checks whether the Google account is already associated with a TORBIDD account. 9. If an existing account is found, the system authenticates the visitor. 10. If no existing account is found, the system creates a new TORBIDD account using the available Google account information. 11. The system creates an authenticated session. 12. The system redirects the visitor to the procurement opportunity dashboard. 13. Personalized features such as bookmarks, comparisons, saved opportunities, and notifications become available.
Postconditions	1. The visitor is successfully authenticated through Google OAuth, or a new TORBIDD account is created. 2. An authenticated TORBIDD session is established. 3. Personalized features become available. 4. The visitor is redirected to the procurement opportunity dashboard.

3.3 Procurement Opportunity Dashboard

3.3.1 Description and Priority

The dashboard serves as the homepage for users, displaying the latest software-related procurement opportunities published by BMA.

Priority: High

3.3.2 Functional Requirements

- POD-1: The system shall display active software-related procurement opportunities from BMA.
- POD-2: The system shall display the following details for each procurement item on the dashboard: Project Title, Issuing Department, Estimated Budget, and Submission Deadline.
- POD-3: The system shall allow users to click on any procurement listing to navigate to its full details page.

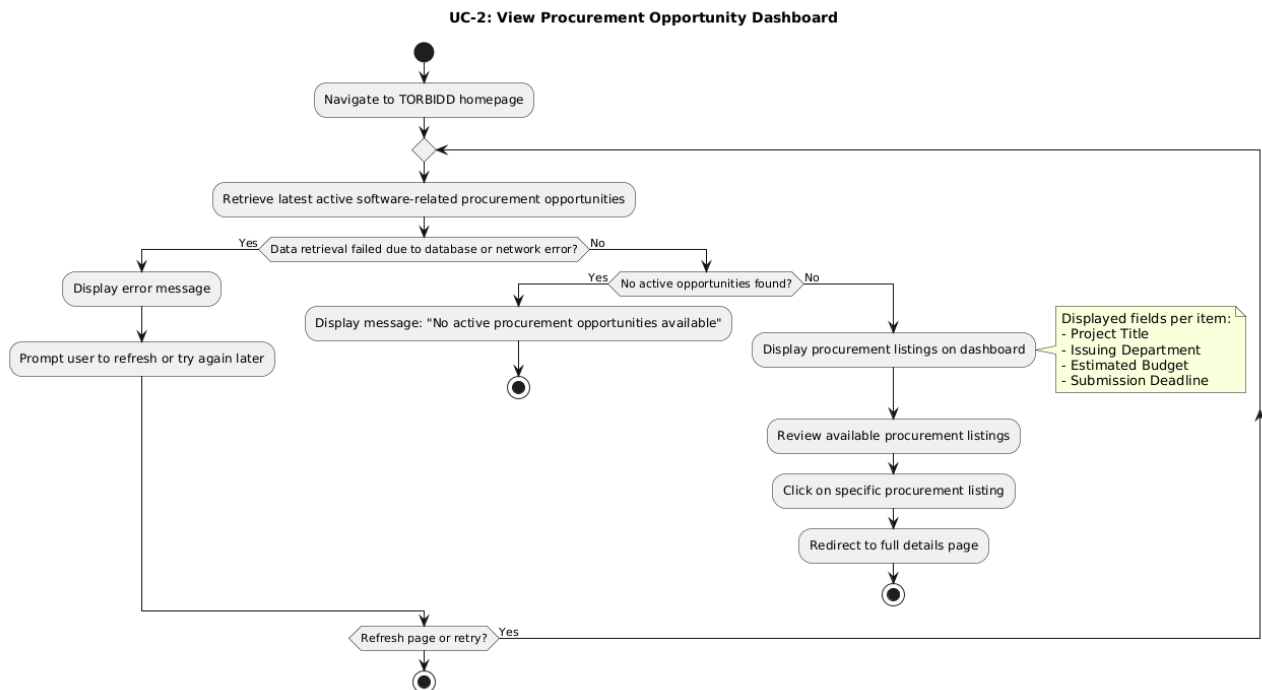
3.3.3 Related Use Cases

UC-2: View Procurement Opportunity Dashboard

Item	Details
Use Case ID	UC-2
Use Case Name	View Procurement Opportunity Dashboard
Use Case Type	Casual
Primary Actor	User / Visitor
Supporting Actor	BMA Procurement Database (or internal data source)
Description	A user accesses the TORBIDD homepage to view the procurement opportunity dashboard. The dashboard displays the latest active, software-related procurement opportunities published by the Bangkok Metropolitan Administration (BMA). The user can review high-level details of each project and select a specific listing to view its complete information.
Preconditions	1. The user has accessed the TORBIDD website. 2. The system has active procurement data available to display.
Main Flow	1. The user navigates to the TORBIDD homepage (Procurement Opportunity Dashboard). 2. The system retrieves the latest active software-related procurement opportunities from BMA. 3. The system displays the retrieved procurement listings on the dashboard. 4. For each procurement item, the system displays the Project Title, Issuing Department, Estimated Budget, and Submission Deadline. 5. The user reviews the available procurement listings on the dashboard. 6. The user selects a procurement listing of interest. 7. The system redirects the user to the full details page for the selected procurement opportunity.
Alternative Flows	A1 No Active Opportunities: If there are no active software-related procurement opportunities currently available from BMA, the system displays a friendly message indicating that no opportunities are currently active. A2 Data Retrieval Failure: If the system is unable to retrieve the procurement opportunities due to a database or network error, the system displays an error message prompting the user to refresh the page or try again later.

Postconditions	1. The user has successfully viewed the active procurement opportunities on the dashboard. 2. If a listing was selected, the user is viewing the full details page of the chosen procurement opportunity.
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User Flow Diagram (View Procurement Opportunity Dashboard):



3.4 Smart Search & Filtering

3.4.1 Description and Priority

Users can efficiently search and filter procurement opportunities to find projects relevant to their expertise.

Priority: High

3.4.2 Functional Requirements

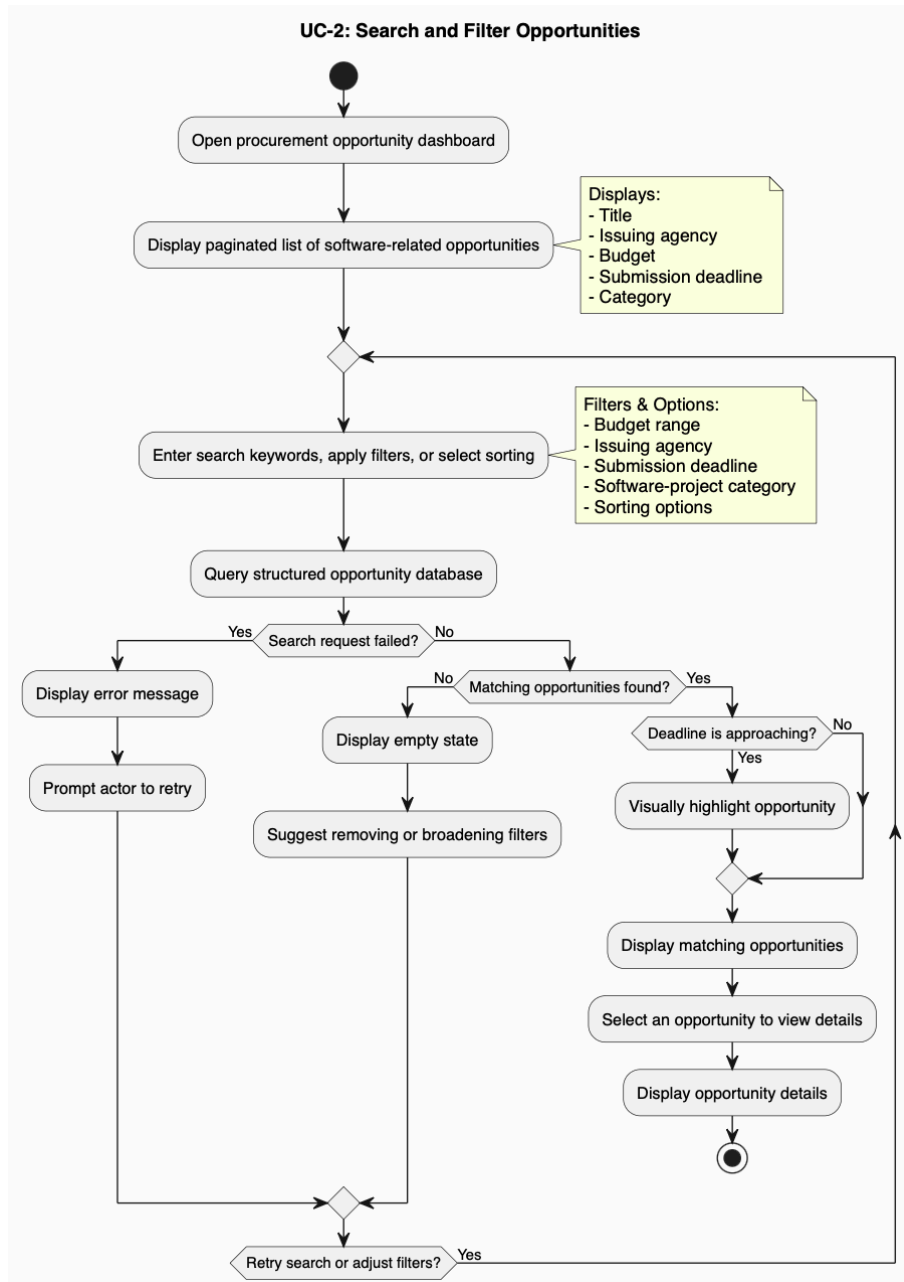
- SSF-1: The system shall allow users to search procurement opportunities using keywords.
- SSF-2: The system shall allow users to filter procurement opportunities by Issuing Department.
- SSF-3: The system shall allow users to filter procurement opportunities by Estimated Budget Range.
- SSF-4: The system shall allow users to filter procurement opportunities by Submission Deadline.
- SSF-5: The system shall allow users to filter procurement opportunities by Project Category.
- SSF-6: The system shall update the dashboard feed dynamically based on the applied search and filter criteria.

3.4.3 Related Use Cases

UC-3: Search and Filter Opportunities

Item	Details
Use Case ID	UC-3
Use Case Name	Search and Filter Opportunities
Primary Actors	Guest, Registered User
Description	The actor searches for software-related government procurement opportunities using keywords and structured filters.
Preconditions	1. The system contains at least one published procurement opportunity. 2. The actor has opened the procurement opportunity dashboard.
Main Flow	1. The actor opens the procurement opportunity dashboard. 2. The system displays a paginated list of current software-related opportunities. 3. Each opportunity displays its title, issuing agency, budget, submission deadline, and category. 4. The actor enters one or more keywords. 5. The actor optionally applies filters such as: a. Budget range b. Issuing agency c. Submission deadline d. Software-project category 6. The actor optionally selects a sorting option. 7. The system queries the structured opportunity database. 8. The system displays the matching opportunities. 9. The actor selects an opportunity to view its details.
Alternative Flows	A1 No Matching Opportunities: If no opportunities match the specified search criteria, the system displays an empty state and suggests removing or broadening one or more filters. A2 Search Request Failure: If the search request fails, the system displays an error message and allows the actor to retry. A3 Approaching Deadline: If an opportunity has an approaching submission deadline, the system visually highlights the opportunity.
Postconditions	1. Matching procurement opportunities are displayed. 2. The selected search and filter criteria remain active until they are cleared or changed. 3. If an opportunity is selected, the actor is able to view its details.

User Flow Diagram (Search and Filter Opportunities):



3.5 TOR Information Extraction

3.5.1 Description and Priority

The system automatically extracts key information from TOR PDF documents and presents it in a structured format.

Priority: High

3.5.2 Functional Requirements

- TIE-1: The system shall automatically extract important information (Project Title, Department, Budget, Project Scope, Submission Deadline, and Bidder Qualifications) from TOR PDF documents.
- TIE-2: The system shall store the extracted data in the database linked to the respective procurement record.

3.5.3 Related Use Cases

UC-4: Extract and Classify TOR Information

Item	Details
Use Case ID	UC-4
Use Case Name	Extract and Classify TOR Information
Primary Actor	Data Collection Service
Supporting Actor	Third-Party AI Service
Description	The system processes a TOR document, extracts structured procurement information, and classifies the procurement listing as a software or non-software project.
Preconditions	1. A new procurement announcement and its TOR document have been collected. 2. The Third-Party AI Service is available.
Main Flow	1. The Data Collection Service submits the TOR document to the AI Service. 2. The AI Service processes the Thai document content. 3. The AI Service extracts structured information, including: a. Budget b. Submission deadline c. Project scope d. Deliverables e. Bidder qualifications f. Contact information 4. The system determines whether the procurement listing is related to a software project. 5. If the listing is software-related, the system assigns an appropriate project category. 6. The system records extraction and classification confidence indicators. 7. The system validates the extracted fields. 8. If the confidence level is acceptable, the system publishes the structured record to the opportunity dashboard.

	9. If the confidence level is low, the system adds the record to the administrator review queue.
Alternative Flows	A1 Invalid or Unreadable TOR Document: If the PDF is corrupted, unreadable, or image-only, the system queues the record for administrator review. A2 AI Service Unavailable: If the AI Service is unavailable, the system records the failure and schedules a retry. A3 Mandatory Fields Not Extracted: If mandatory fields cannot be extracted, the system marks the affected fields as incomplete. A4 Non-Software Classification: If the listing is classified as a non-software project, the system prevents it from appearing on the public software-opportunity dashboard. A5 Duplicate Record Detected: If a duplicate procurement record is detected, the system links it to the existing record or rejects the duplicate according to the system's duplicate-handling rules.
Postconditions	1. A structured procurement opportunity record is created. 2. The record is either published to the public dashboard, excluded as a non-software project, or queued for administrator review.

User Flow Diagram (Extract and Classify TOR Information):



3.6 Software Project Classification

3.6.1 Description and Priority

The system automatically identifies whether a procurement project is software-related.

Priority: High

3.6.2 Functional Requirements

- SPC-1: The system shall automatically classify whether a procurement is software-related.
- SPC-2: The system shall classify a project as software-related if it matches one or more of the following criteria: Website development, Mobile application, Information systems, AI solutions, Database systems, or Software maintenance.

3.7 Procurement Detail Viewer

3.7.1 Description and Priority

Users can view complete procurement information extracted from the TOR.

Priority: High

3.7.2 Functional Requirements

- PDV-1: The system shall display detailed information for every procurement opportunity.
- PDV-2: The system shall display the structured data extracted from the TOR, including: Project Title, Budget, Department, Project Description, Scope of Work, Bidder Qualifications, and Submission Deadline.
- PDV-3: The system shall provide a link or embedded viewer for users to access and download the original TOR PDF document.

3.7.3 Related Use Cases

UC-5: View Opportunity Details

Item	Details
Use Case ID	UC-5
Use Case Name	View Opportunity Details
Primary Actor	Guest, Registered User
Description	The actor views structured information extracted from a selected TOR document without having to manually read the complete PDF.
Preconditions	1. The selected opportunity has been collected and classified as software-related. 2. The opportunity is published on the dashboard.

Main Flow	1. The actor selects an opportunity from the dashboard or search results. 2. The system retrieves the structured opportunity record. 3. The system displays: a. Project title b. Issuing government agency c. Procurement budget d. Submission deadline e. Project scope f. Expected deliverables g. Bidder qualifications h. Contact information i. Project category j. Extraction or classification confidence 4. The system displays a link to the original announcement or TOR PDF. 5. The actor reviews the information. 6. A registered user may proceed to bookmark, compare, or configure an alert for the opportunity.
Alternative Flow	A1 Low Extraction Confidence: If extraction confidence is low, the system displays a warning that some information may be incomplete. A2 Unextractable Field: If a field could not be extracted, the system marks it as unavailable. A3 Original TOR Document Unavailable: If the original TOR document is unavailable, the system disables the source link and displays a notice. A4 Expired Opportunity: If the opportunity has expired, the system displays its closed or expired status.
Postconditions	1. Notification preferences are stored in the user's account. 2. The user receives notifications for future matching opportunities.

3.8 Opportunity Notification

3.8.1 Description and Priority

The system notifies users when new procurement opportunities match their interests.

Priority: Medium

3.8.2 Functional Requirements

- ON-1: The system shall notify users when a new procurement matches their interests.
- ON-2: The system shall allow users to define, update, or toggle their notification criteria and delivery preferences.

3.8.3 Related Use Cases

UC-6: Configure and Receive Opportunity Alerts

Item	Details
Use Case ID	UC-6
Use Case Name	Configure and Receive Opportunity Alerts
Primary Actor	Registered User
Supporting Actor	Data Collection Service, Email Delivery Service
Description	A registered user configures matching criteria and receives in-app or email notifications when new procurement opportunities match those criteria.
Preconditions	1. The user is authenticated. 2. The user has a valid email address.
Main Flow	1. The user opens notification preferences. 2. The system displays the available notification criteria. 3. The user configures one or more of the following: a. Keywords b. Software-project categories c. Budget range d. Issuing agencies 4. The user selects notification channels: a. In-app notification b. Email notification c. Both channels 5. The user saves the preferences. 6. The Data Collection Service detects and classifies a new opportunity. 7. The system compares the opportunity with saved user preferences. 8. If a match is found, the system creates an in-app notification. 9. If email notification is enabled, the Email Delivery Service sends an email. 10. The user opens the notification and accesses the matching opportunity.

Postconditions	1. Notification preferences are stored in the user's account. 2. The user receives notifications for future matching opportunities.
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3.9 Historical Price Dashboard

3.9.1 Description and Priority

Displays historical software procurement data for comparison.

Priority: High

3.9.2 Functional Requirements

- HPD-1: The system shall display historical software procurement projects for comparison.
- HPD-2: The system shall allow users to filter the historical data by Year.
- HPD-3: The system shall allow users to filter the historical data by Department.
- HPD-4: The system shall allow users to filter the historical data by Project Category.

3.9.3 Related Use Cases

UC-7: View Historical Budget and Outlier Analysis

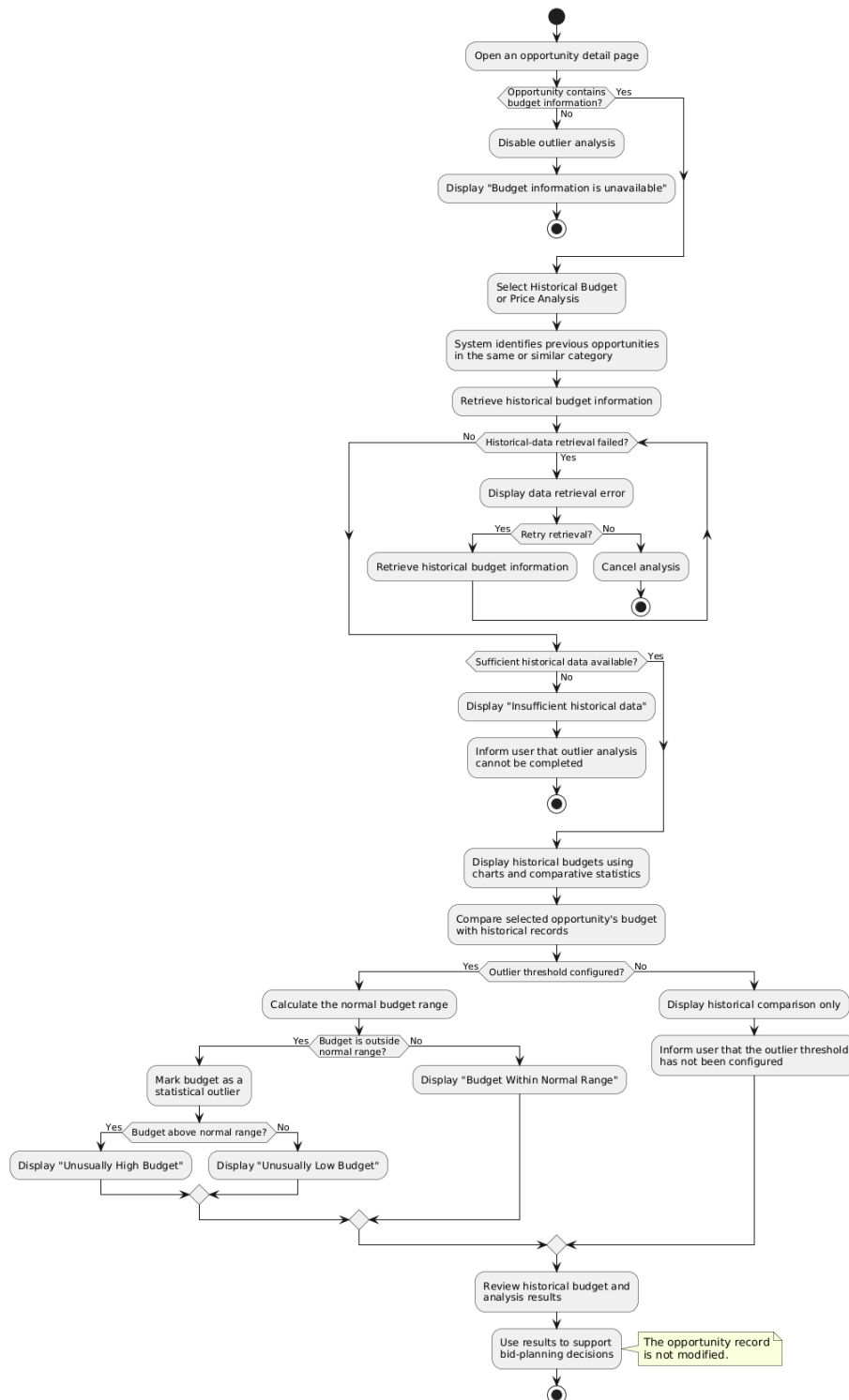
Item	Details
Use Case ID	UC-7
Use Case Name	View Historical Budget and Outlier Analysis
Primary Actor	Registered User
Description	The user reviews historical procurement budgets for similar projects and checks whether the selected opportunity's budget is unusually high or low.
Preconditions	1. The selected opportunity contains budget and category information. 2. Historical comparable opportunities are available.

Main Flow	1. The user opens an opportunity detail page. 2. The user selects the historical budget or price-analysis option. 3. The system identifies previous opportunities in the same or similar category. 4. The system retrieves historical budget information. 5. The system displays the information using charts or comparative statistics. 6. The system compares the selected opportunity's budget with historical records. 7. The system indicates whether the budget is within the normal range or is a statistical outlier. 8. The user reviews the result to support bid-planning decisions.
Postconditions	1. Historical budget information is displayed. 2. The opportunity record is not modified.

Note: The exact statistical method and outlier threshold must be confirmed with the client.

User Flow Diagram (View Historical Budget and Outlier Analysis):

UC-6: View Historical Budget and Outlier Analysis



3.10 Budget Outlier Analysis

3.10.1 Description and Priority

The system analyzes whether a procurement budget appears significantly different from similar historical software projects.

Priority: High

3.10.2 Functional Requirements

- BOA-1: The system shall compare the current procurement budget with historical software procurements.
- BOA-2: The system shall generate and display a visual warning if the current project's budget is significantly higher or lower than the historical average.

3.11 Bookmark Management

3.11.1 Description and Priority

Users can save procurement opportunities for later review.

Priority: Low

3.11.2 Functional Requirements

- BM-1: The system shall allow users to bookmark procurement opportunities.
- BM-2: The system shall allow users to remove a bookmark from a previously saved opportunity.

3.12 User Profile

3.12.1 Description and Priority

Users manage their account information and notification preferences.

Priority: Medium

3.12.2 Functional Requirements

- UP-1: The system shall allow users to manage their account.
- UP-2: The system shall allow users to securely change their account password.
- UP-3: The system shall allow users to configure their notification preferences.

3.12.3 Related Use Cases

UC-8: Manage User Profile and Preferences

Item	Details
Use Case ID	UC-8
Use Case Name	Manage User Profile and Preferences

Primary Actor	Registered User
Supporting Actor	-
Description	A registered user views and updates account information and notification preferences.
Preconditions	The user is authenticated.
Main Flow	1. The user opens the profile page. 2. The system displays the user's account information. 3. The user edits permitted fields such as: 4. Name 5. Organization 6. Organization or professional role 7. Notification preferences 8. The user saves the changes. 9. The system validates the submitted information. 10. The system updates the user profile. 11. The system displays confirmation. 12. Postconditions: 13. The user profile and preferences are updated. 14. The new settings apply to future system interactions and notifications.

3.13 Administration Panel

3.13.1 Description and Priority

Allows administrators to manage the platform.

Priority: Medium

3.13.2 Functional Requirements

- AP-1: The system shall provide a secure, role-restricted Administration Panel accessible only to authorized admin users.
- AP-2: The system shall allow admins to monitor the status of automatic procurement data ingestion.
- AP-3: The system shall allow admins to view, edit, or manually correct data extracted from TOR documents.
- AP-4: The system shall allow admins to override or correct the automated Software Project Classification.
- AP-5: The system shall allow admins to view system processing logs to troubleshoot extraction or classification errors.
- AP-6: The system shall allow admins to manage user accounts, including viewing, suspending, or deleting users.

3.13.3 Related Use Cases

UC-9: Manage Users

Item	Details
Use Case ID	UC-9
Use Case Name	Manage Users
Primary Actor	System Administrator
Supporting Actor	-
Description	An administrator manages TORBIDD user accounts and access roles.
Preconditions	1. The administrator is authenticated. 2. The administrator has user-management permission.
Main Flow	1. The administrator opens the user-management page. 2. The system displays registered user accounts. 3. The administrator searches for or selects a user. 4. The system displays the account's basic information, role, and status. 5. The administrator performs an authorized action, such as: 6. Change the user's role 7. Activate an account 8. Suspend an account 9. Restore a suspended account 10. The system validates the change. 11. The system updates the user account. 12. The system records the administrative action.
Alternative Flow	A1 Unauthorized Role Change: If the administrator attempts an unauthorized role change, the system denies the operation. A2 Account Not Found: If the account cannot be found, the system displays an error. A3 Active Sessions During Suspension: If the account has active sessions when suspended, the system terminates those sessions. A4 Administrator Cancels Changes: The administrator may cancel before saving changes.
Postconditions	1. The selected user account reflects the updated role or status. 2. The administrative action is recorded.

4. Non-functional Requirements

This section specifies the quality attributes, performance expectations, security requirements, and operational constraints of TORBIDD.

4.1 Performance Requirements

- **NFR-4.1.1** Dashboard search and filter results shall be returned within 3 seconds under normal operating conditions. **(High)**
- **NFR-4.1.2** Newly published procurement announcements shall be collected and made available on the dashboard within 24 hours of publication, subject to source availability. **(Medium)**
- **NFR-4.1.3** Procurement detail pages shall load within 3 seconds under normal operating conditions, excluding the time required to retrieve external documents. **(Medium)**

4.2 Security Requirements

- **NFR-4.2.1** All client-server communication shall use HTTPS/TLS. **(High)**
- **NFR-4.2.2** The system shall restrict administrative functions to authenticated administrator accounts. **(High)**
- **NFR-4.2.3** Users shall only be able to access and modify their own personal data associated with their own accounts. **(High)**
- **NFR-4.2.4** The system shall enforce authentication and authorization mechanisms for all protected functions and data. **(High)**

4.3 Usability Requirements

- **NFR-4.3.1** The dashboard, search, and procurement detail views shall be usable on desktop and mobile screen sizes. **(High)**
- **NFR-4.3.2** Procurement information extracted from TOR documents shall be presented in a clear and structured format. **(High)**
- **NFR-4.3.3** The system interface shall support Thai content where applicable. **(Medium)**
- **NFR-4.3.4** Error and status messages shall be presented in clear and understandable language. **(Medium)**

4.4 Reliability Requirements

- **NFR-4.4.1** The data collection service shall retry a failed source request at least once before marking the source as requiring administrator attention. **(Medium)**
- **NFR-4.4.2** The dashboard shall remain available using previously stored procurement data if the AI extraction service is temporarily unavailable. **(High)**
- **NFR-4.4.3** Temporary failure of one government data source shall not prevent the system from processing data from other available sources. **(Medium)**

4.5 Maintainability Requirements

- **NFR-4.5.1** Data collection logic for each government source shall be modular so that sources can be added or modified without significantly affecting unrelated collection components. **(Medium)**
- **NFR-4.5.2** The system shall use a modular architecture that allows AI processing, data collection, search, and notification components to be modified independently where practical. **(Medium)**

4.6 Portability Requirements

- **NFR-4.6.1** The application shall be deployable using Docker on a compatible cloud-hosting environment. **(High)**
- **NFR-4.6.2** The web application shall support modern versions of commonly used browsers, including Google Chrome, Mozilla Firefox and Safari. **(Medium)**

4.7 Scalability Requirements

- **NFR-4.7.1** The database and application architecture shall support the addition of further government agencies without requiring fundamental redesign of the system. **(Medium)**
- **NFR-4.7.2** The system shall support growth in the number of procurement records and users without requiring fundamental changes to the core architecture. **(Medium)**

5. Other Requirements

5.1 Legal and Compliance Requirements

- **LEGAL-1:** The system shall use procurement information obtained from publicly available government sources and shall not modify or submit procurement information to government procurement systems.
- **LEGAL-2:** The system shall handle bidder, agency, and contact information in accordance with applicable data-protection and privacy requirements when such information is collected or stored.
- **LEGAL-3:** The system shall not intentionally collect or retain personally identifiable bidder information beyond what is necessary to support its intended procurement-information functions.
- **LEGAL-4:** The system shall use third-party services, APIs, and external data sources in accordance with their applicable terms of service, usage policies, and access conditions.

5.2 Localization Requirements

- **LOC-1:** The system shall support **Thai procurement content**, including TOR documents and extracted information.
- **LOC-2:** The user interface shall correctly display Thai text. The final user-interface language configuration shall be confirmed during detailed UI design.

5.3 Out-of-Scope Features

The following features are outside the scope of the current TORBIDD implementation:

- Submission of bids or procurement documents on behalf of users
- Modification of information on government procurement systems
- Direct authentication with government procurement systems
- Private or restricted procurement data that is not publicly available
- Automated bidding or bid decision-making
- Payment processing or monetization features
- Native mobile applications
- Advanced AI model training or active learning from administrator corrections
- Integration with additional government sources beyond the agreed initial scope without further development

5.4 Development and Validation

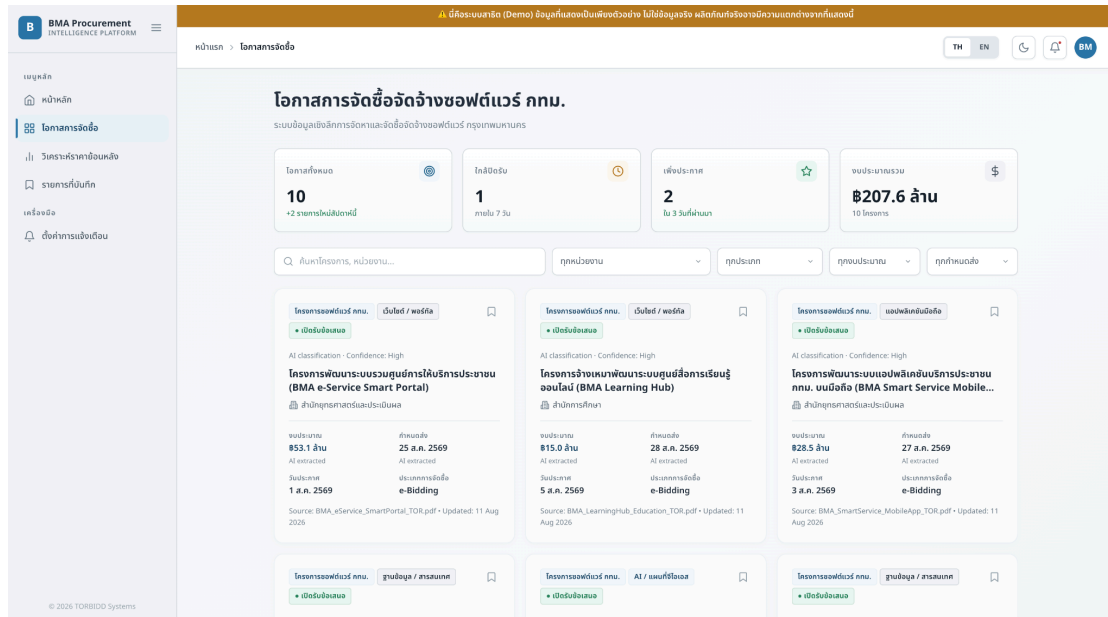
- **DEV-1:** Testing and validation shall be performed by the project team throughout development.
- **DEV-2:** The system shall be validated against the functional and non-functional requirements specified in this SRS before final delivery.

6. Additional Material

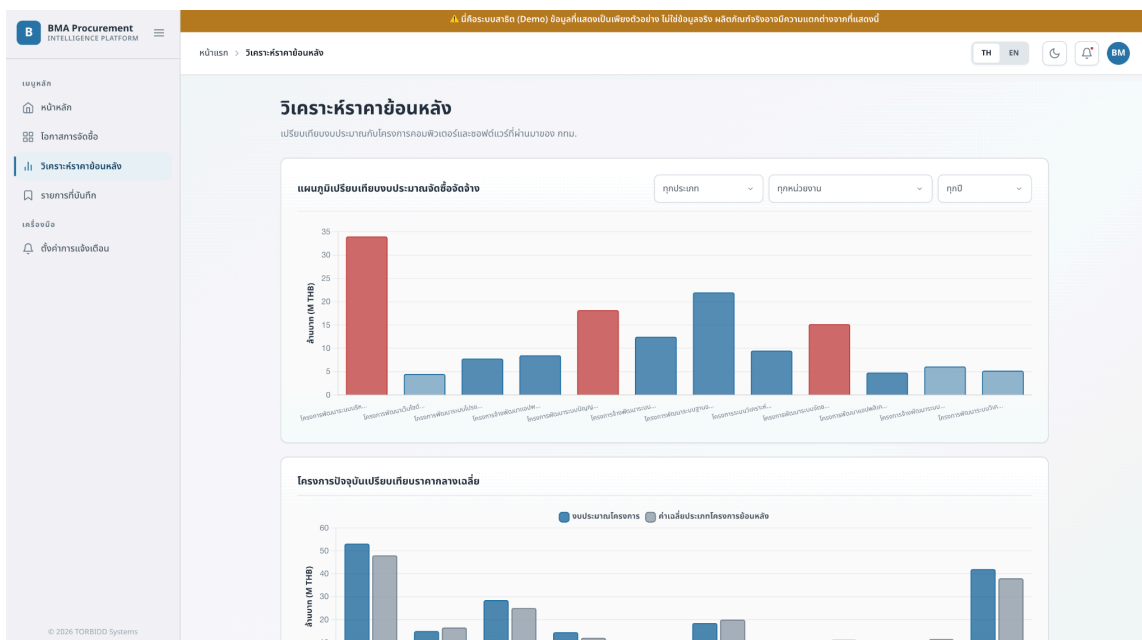
6.1 Website Preview



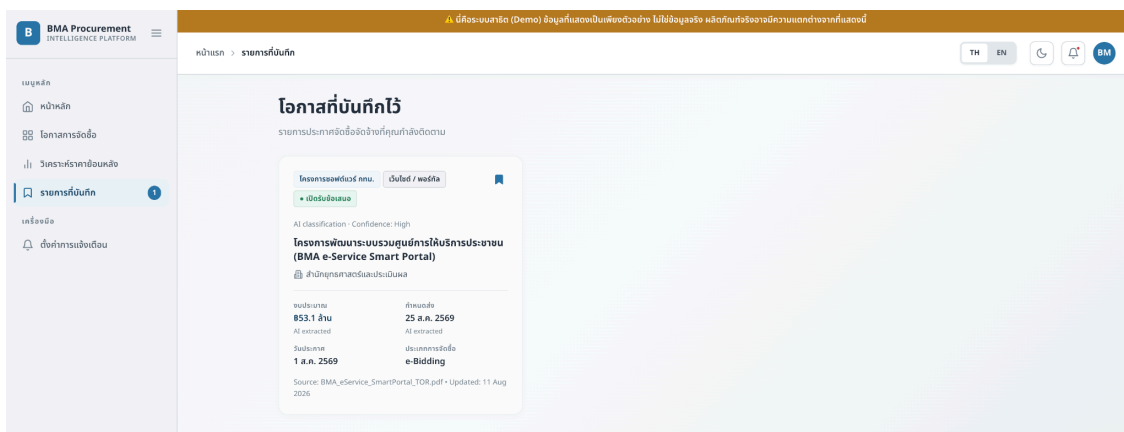
Home page



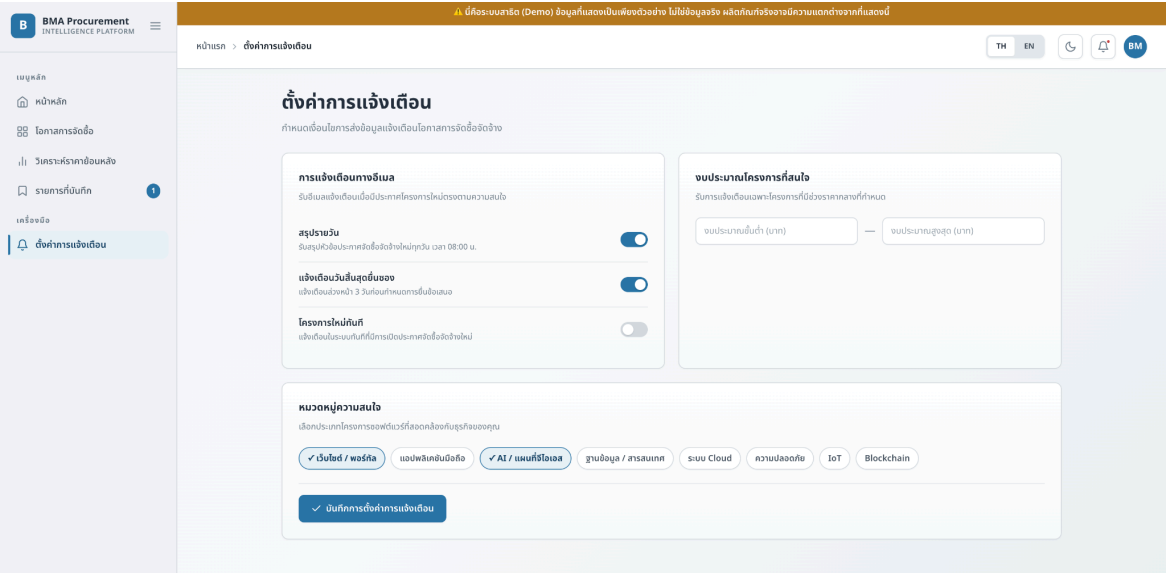
Opportunities board page



Price Analysis page



Saved Items page



Notification Setting Page

Appendix A: Glossary

Term	Definition
TOR	Terms of Reference, the official document an agency publishes describing a procurement's scope, budget, deliverables, and bidder qualifications.
BMA	Bangkok Metropolitan Administration, the primary initial data source for procurement announcements.
TORBIDD	The name of the platform being specified in this document.
Extraction confidence	A system-generated indicator of how reliable an AI-extracted field is, used to route low-confidence records to administrator review.
Outlier analysis	Statistical comparison flagging an opportunity's budget as unusually high or low relative to similar historical opportunities.
Vertex AI	The AI platform is referenced in the proposal as the intended engine for TOR PDF extraction and classification.
MongoDB Atlas	The cloud-hosted database referenced in the proposal for storing structured opportunity data.

Appendix B: Requirements Analysis Notes

This section documents unresolved issues, ambiguities, and justified assumptions identified during the requirements analysis process. These notes identify areas where the proposal or client requirements were incomplete, high-level, or required interpretation.

Open Issues

The following issues may require clarification during further requirements analysis or future development

- **OI-1: Government Source Scope**, The exact list of government agencies to be monitored beyond the sources identified in Section 2.6 is not fully finalized. The proposal indicates coverage of up to five government agencies, subject to confirmation.
- **OI-2: Classification Correction Feedback**, It has not been determined whether administrator corrections to AI-generated classifications and extracted information should be used to improve future AI processing or remain as manual overrides.
- **OI-3: Procurement Data Retention**, The retention policy for expired, cancelled, or closed procurement opportunities has not been specified. It remains to be determined whether such records should be archived or permanently deleted.

Unclear or Ambiguous Requirements

The following requirements require further clarification or a defined implementation approach

- **UR-1: Budget Outlier Analysis**, The proposal specifies budget outlier analysis but does not define the statistical method or threshold to be used. A specific method, such as percentile-based thresholds or standard-deviation analysis, should be defined before implementation.
- **UR-2: User Profile Scope**, The exact scope of the user profile feature is not fully defined. It is unclear whether the profile should include only basic user information or additional preferences such as technology interests, saved searches, or organization information.

Justified Assumptions

Where the proposal does not specify sufficient implementation detail, the following assumptions are made to allow the requirements to be defined and tested

- **JA-1: Notification Channels**, Notifications will support both in-app/web notifications and email notifications.
- **JA-2: Administrator Functions**, The administration panel will provide, at minimum, user management, procurement-data review, classification correction, and data-quality management functions.
- **JA-3: Saved Opportunities**, Users will be able to save procurement opportunities to a personal list and later remove them from that list.
- **JA-4: Historical Data Retention**, Historical procurement records will be retained for analysis and comparison unless a future data-retention policy specifies otherwise.
- **JA-5: AI Processing Review**, AI-extracted information and classifications may require administrator review or correction when the system identifies errors or insufficient confidence.

Appendix C: Traceable Matrix

Feature	Functional Requirement	Use Case
User Authentication and Access Control	AUTH-1, AUTH-2, AUTH-3	UC-1
Procurement Opportunity Dashboard	POD-1, POD-2, POD-3	UC-2

Smart Search & Filtering	SSF-1, SSF-2, SSF-3, SSF-4, SSF-5, SSF-6	UC-3
TOR Information Extraction	TIE-1, TIE-2	UC-4
Software Project Classification	SPC-1, SPC-2	-
Procurement Detail Viewer	PDV-1, PDV-2, PDV-3	UC-5
Opportunity Notification	ON-1, ON-2	UC-6
Historical Price Dashboard	HPD-1, HPD-2, HPD-3, HPD-4	UC-7
Budget Outlier Analysis	BOA-1, BOA-2	-
Bookmark Management	BM-1, BM-2	-
User Profile	UP-1, UP-2, UP-3	UC-8
Administration Panel	AP-1, AP-2, AP-3, AP-4, AP-5, AP-6	UC-9